English

(1015CMD303361230030)

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

NEET (UG)
MINOR
18-06-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

Test Booklet Code

This Booklet contains **32** pages.

E4

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II). 50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720**.
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : ________________

Form Number : in figures ________________

: in words ________________

Centre of Examination (in Capitals) : ________________

Candidate's Signature : ____________ **Invigilator's Signature :** ____________

Your Target is to secure Good Rank in Pre-Medical 2024

Topic : LENSES, OPTICAL INSTRUMENTS AND EYE DEFECTS, WAVE OPTICS, ELECTROSTATICS, CURRENT, CAPACITOR [BACK UNIT] :- ELECTROSTATICS

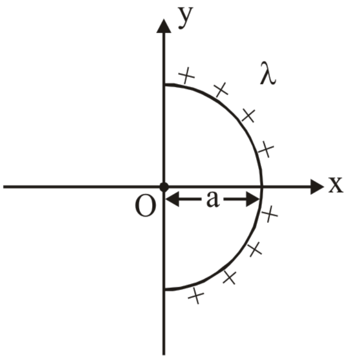
SECTION-A (PHYSICS)

1. The total electric flux emanating from a closed surface enclosing an α-particle (e = electronic charge) is :-

(1) $\frac{2e}{\varepsilon_0}$ (2) $\frac{e}{\varepsilon_0}$

(3) $e\,\varepsilon_0$ (4) $\frac{\varepsilon_0 e}{4}$

2.



Semi circular ring of radius a is uniformly charged with linear charge density λ. Find out electric field intensity at point O.

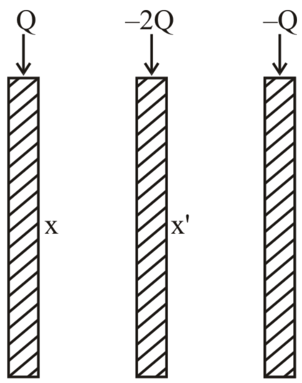
(1) $\frac{\lambda}{2\pi\varepsilon_0 a}(-\hat{i})$ (2) $\frac{\lambda}{4\pi\varepsilon_0 a}\hat{i}$

(3) $\frac{\lambda}{2\pi\varepsilon_0 a}(+\hat{i})$ (4) $\frac{\lambda}{4\pi\varepsilon_0 a}(-\hat{i})$

3. The force of attraction between two co-axial dielectric dipoles whose centres are "r" meter apart varies with distance as :-

(1) r^{-1} (2) r^{-2} (3) r^{-3} (4) r^{-4}

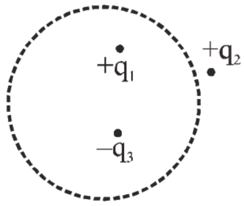
4. Charge on surface x & x' is :-



(1) Q, –2Q (2) –2Q, 0

(3) 2Q, 0 (4) 0, –2Q

5. Consider the charge configuration and spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface the electric field will be due to



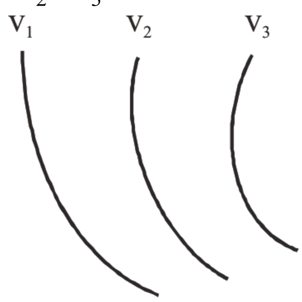
(1) q_2

(2) Only the positive charges

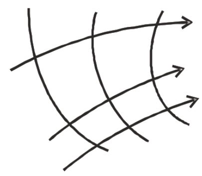
(3) All the charges

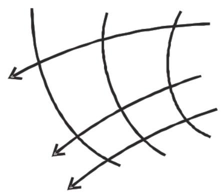
(4) $+q_1$ and $-q_3$

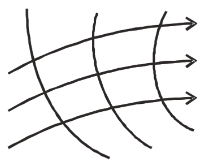
6. Some equipotential surfaces are shown in the figure such that $V_1 < V_2 < V_3$ in an electric field region.

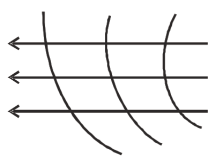


Choose corresponding electric field lines present in this region.

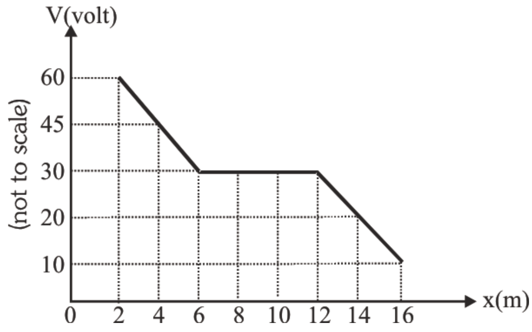
(1) 

(2) 

(3) 

(4) 

7. The variation of potential with distance x from a fixed point is shown in figure. The electric field at x =13 m is



(1) 7.5 volt/meter (2) −7.5 volt/meter

(3) 5 volt/meter (4) −5 volt/meter

8. Electric potential is given by :

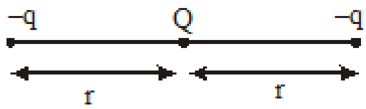
$V = 6x - 8xy^2 - 8y + 6yz - 4z^2$

Electric field at the origin is -

(1) $-6\hat{i} + 8\hat{j}$ (2) $6\hat{i} - 8\hat{j}$

(3) $\hat{i} + \hat{j}$ (4) Zero

9. Three charges are placed as shown in fig if the electric potential energy of system is zero, then Q : q is:-



(1) $\frac{Q}{q} = \frac{-2}{1}$

(2) $\frac{Q}{q} = \frac{2}{1}$

(3) $\frac{Q}{q} = \frac{-1}{2}$

(4) $\frac{Q}{q} = \frac{1}{4}$

10. n small drops of same size are charged to 'V' volt each. If they coalesce to form a single large drop, then its potential will be -

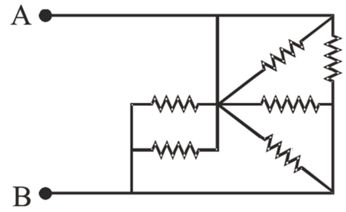
(1) V/n

(2) Vn

(3) $Vn^{1/3}$

(4) $Vn^{2/3}$

11. The circuit shown has six resistors of equal resistance R. Find the equivalent resistance between A and B :



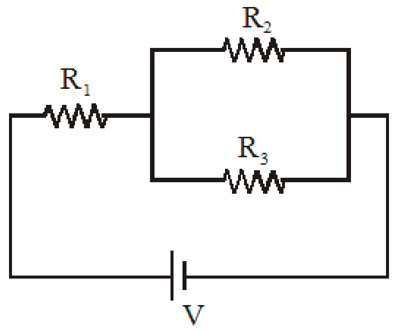
(1) $\frac{11R}{12}$

(2) $\frac{13R}{12}$

(3) $\frac{R}{5}$

(4) $\frac{15R}{12}$

12. Three identical resistors $R_1 = R_2 = R_3$ are connected as shown to a battery of constant e.m.f. The power dissipated is :-



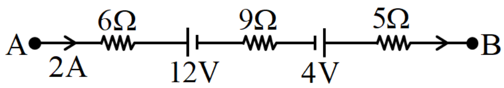
(1) The least in R_1

(2) Greatest in R_1

(3) In the ratio 1 : 2 resistance R_1 and R_2 respectively

(4) The same in R_1 and in the parallel combination of R_2 and R_3

13. The potential difference between A and B in the following figure is



(1) 32 V

(2) 48 V

(3) 24 V

(4) 14 V

14. Charge passing through a conductor of cross-section area $A = 0.3\ m^2$ is given by $q = 3t^2 + 5t + 2$ in coulomb, where t is in second. What is the value of drift velocity at t = 2s ?
(Given, $n = 2 \times 10^{25}/m^3$)

(1) 0.77×10^{-5} m/s (2) 1.77×10^{-5} m/s

(3) 2.08×10^{5} m/s (4) 0.57×10^{-5} m/s

15. The capacitance C of a capacitor is :-

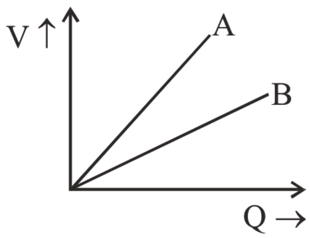
(1) Independent of the charge and potential of the capacitor.

(2) Dependent on the charge and independent of potential.

(3) Independent of the geometrical configuration of the capacitor.

(4) Independent of the dielectric medium between the two conducting surfaces of the capacitor

16. The graph shows the variation of voltage V across the plates of two capacitors A and B with charge 'Q'. Relation between capacitance of A and B is :



(1) $C_A > C_B$ (2) $C_A = C_B$

(3) $C_A < C_B$ (4) None of these

17. If the distance between the plates of a capacitor of capacitance C is made half and the area of plates is doubled then what will be the capacitance.

(1) 4C (2) 2C

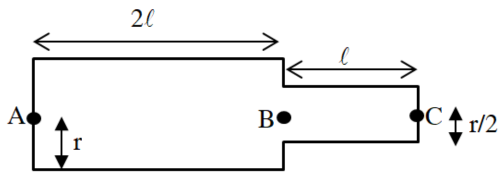
(3) C/2 (4) 8C

18. The inner and outer radii of a spherical condenser are 50 cm and 60cm respectively and a dielectric of constant 6 is filled in it. Its capacity will be:-

(1) 2×10^{-9} farad (2) 4×10^{-9} farad

(3) 6×10^{-10} farad (4) 5×10^{-10} farad

19. A current I flows through a cylindrical conductor. Then the ratio of potential difference along AB and BC is



(1) 1 : 1 (2) 1 : 2

(3) 1 : 4 (4) 2 : 1

20. In a potentiometer, the wire has specific resistance ρ, area of cross-section A and current I is flowing in it. The potential gradient in the wire is

(1) $\frac{IA}{\rho}$ (2) $\frac{\rho A}{I}$

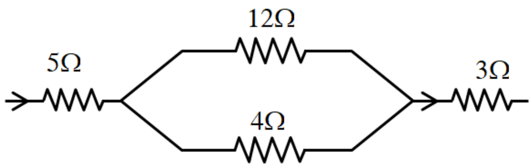
(3) $\frac{A}{\rho I}$ (4) $\frac{I\rho}{A}$

21. A 100 V voltmeter whose resistance is 40 kΩ is connected in series with a very high resistance R. When it is joined across a line of 110 V, it reads 10 V. What is the magnitude of resistance R ?

(1) 440 kΩ (2) 400 kΩ

(3) 360 kΩ (4) None

22.



If the power dissipated in the 12Ω resistance is 12W, then the power dissipated in 3Ω will be :

(1) 12 W (2) 3 W

(3) 8 W (4) 48 W

23. A compound microscope has an eye piece of focal length 10 cm and an objective of focal length 4 cm. Calculate the magnification, if an object is kept at a distance of 5 cm from the objective so that final image is formed at the least distance vision (20 cm) :-

(1) 12 (2) 11 (3) 10 (4) 13

24. A simple telescope consisting of an objective lens of focal length 60 cm and an eye lens of focal length 5 cm, is focused on a distant object. If the object subtends an angle of 2° at the objective lens, the angular width of the image is

(1) 50°

(2) $\left(\frac{1}{6}\right)^{o}$

(3) 10°

(4) 24°

25. An astronomical telescope has an objective lens and eye piece of focal length 50 cm and 5 cm respectively. If distance between lenses is 75 cm then find the distance of object from objective lens for normal adjustment.

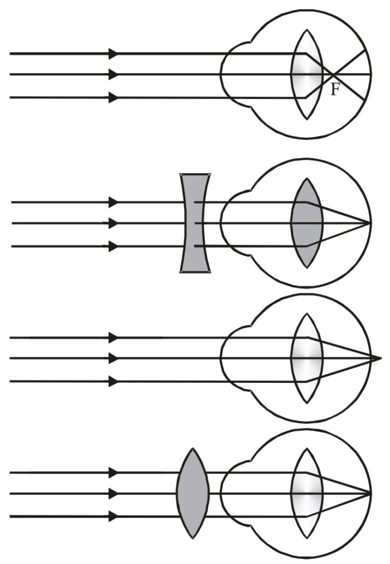
(1) 175 cm

(2) 350 cm

(3) 70 cm

(4) 150 cm

26.



Identify the wrong description of the above figures :-

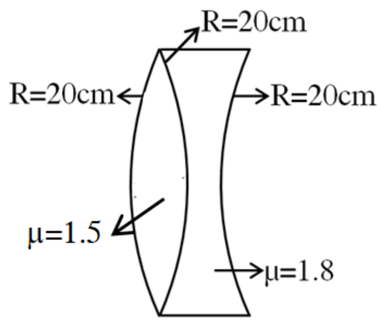
(1) 1 represents far-sightedness

(2) 2 correction for short sightedness

(3) 3 represents far sightedness

(4) 4 correction for far-sightedness

27. The power of the combination ?



(1) +3D

(2) −3D

(3) 5D

(4) −5D

28. A plano convex lens of radius 30 cm and refractive index 1.5 is polished from convex side where should an object be placed from the lens so that real image of same size is formed

(1) 10 cm

(2) 20 cm

(3) 30 cm

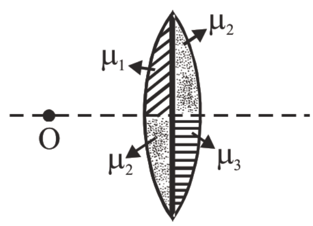
(4) 40 cm

29. A diverging lens of refractive index 1.5 and of focal length 15 cm in air has the same radii of curvature for both sides. If it is immersed in a liquid of refractive index 1.7, calculate the focal length of the lens in the liquid.

(1) 63.75 cm (2) 60 cm

(3) 50 cm (4) 40 cm

30. Number of images formed by lens will be :-



(1) 1 (2) 2 (3) 3 (4) 4

31. The resultant amplitude of a vibrating particle by the super position of two waves y_1=a sin (ωt + π/3) ; y_2 = a sin ωt

(1) a

(2) $\sqrt{2}$a

(3) 2a

(4) $\sqrt{3}$a

32. In YDSE using λ = 6000Å, distance between screen & source is 1m. If the fringe width on the screen is 0.06 cm, the distance between the two coherent sources is :-

(1) 0.01 mm

(2) 1 cm

(3) 1 mm

(4) None

33. In YDSE intensity at a point is $1/4^{th}$ of maximum intensity. Angular position of this point is :-

(1) $\sin^{-1}(\lambda/d)$

(2) $\sin^{-1}(\lambda/2d)$

(3) $\sin^{-1}(\lambda/3d)$

(4) $\sin^{-1}(\lambda/4d)$

34. Three waves of equal frequency having amplitudes 10μm, 4μm 7μm arrive at a given point with successive phase difference of π/2, the amplitude of resultant wave is given by :- (in μm)

(1) 4 (2) 5

(3) 6 (4) 7

35. In YDSE, the intensity of light at a point on the screen where path difference is λ is K, ($\lambda \longrightarrow$ wave length) the intensity at a point where path difference is λ/4, will be :-

(1) K (2) K/4

(3) K/2 (4) Zero

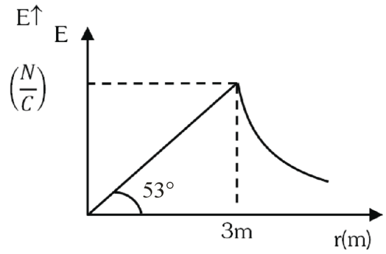
SECTION-B (PHYSICS)

36. Two isolated charged conducting spheres of radii a and b produce the same electric field near their surfaces. The ratio of electric potentials on their surfaces is :-

(1) $\frac{a}{b}$ (2) $\frac{b}{a}$

(3) $\frac{a^2}{b^2}$ (4) $\frac{b^2}{a^2}$

37. For a solid non – conducting sphere, electric field variation with distance from the centre is shown. Then net charge inside the sphere is –



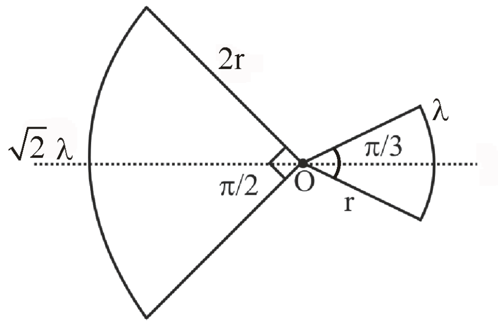
(1) 180 $\pi \in_0$

(2) 162 $\pi \in_0$

(3) 144 $\pi \in_0$

(4) 72$\pi \in_0$

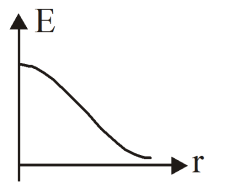
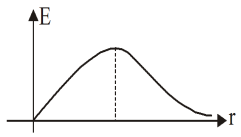
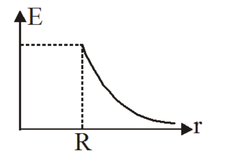
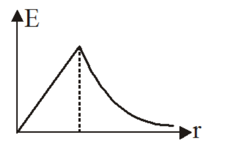
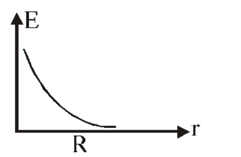
38. Two arc of radius r and 2r carries charge density λ and $\sqrt{2}\,\lambda$ and subtends angle π/3 and π/2 respectively at centre as shown in figure. The electric field at centre is :-



(1) $\frac{\lambda}{4\pi\varepsilon_0 r}$ (2) $\frac{\sqrt{2}\lambda}{\pi\varepsilon_0 r}$

(3) Zero (4) $\frac{2\lambda}{\pi^2\varepsilon_0 r^2}$

39. Column II corresponds to the graph of magnitude of electric field versus distance from centre of charge distribution in Column I.

Column-I		Column-II	
(A)	Ring along its axis	(P)	
(B)	Uniformly charged solid sphere	(Q)	
(C)	Uniformly charged spherical shell	(R)	
(D)	Combination of charge +Q and −Q at the perpendicular bisector	(S)	
		(T)	

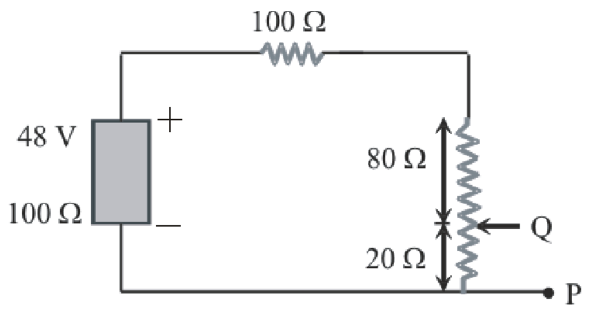
(1) A⟶P, B⟶Q, C⟶R, D⟶S

(2) A⟶S, B⟶Q, C⟶P, D⟶R

(3) A⟶Q, B⟶S, C⟶R, D⟶T

(4) A⟶Q, B⟶S, C⟶R, D⟶P

40. In the circuit, the potential difference across PQ will be nearest to



(1) 9.6 V (2) 6.6 V (3) 4.8 V (4) 3.2 V

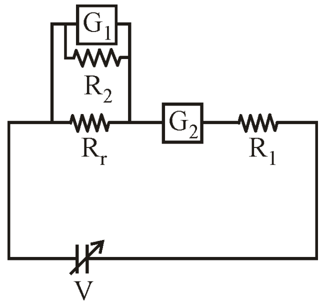
41. An ammeter is obtained by shunting a 30 Ω galvanometer with a 30 Ω resistance. What additional shunt should be connected across it to double its range?

(1) 10 Ω (2) 15 Ω

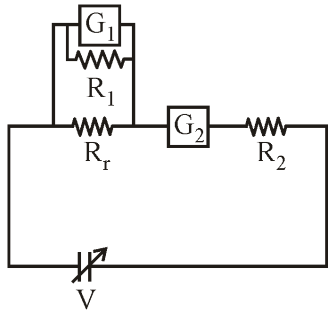
(3) 30 Ω (4) None of these

42. To verify Ohm's law, a student is provided with a test resistor R_T, a high resistance R_1, a small resistance R_2, two identical galvanometers G_1 and G_2, and a variable voltage source V. The correct circuit to carry out the experiment is :-

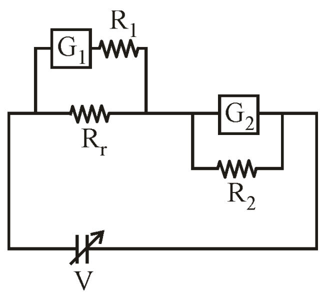
(1)



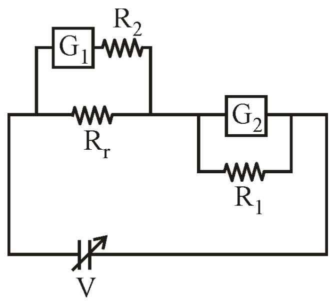
(2)



(3)



(4)

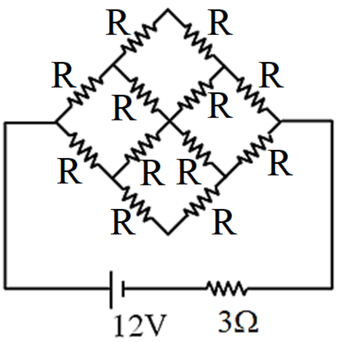


43. A parallel plate capacitor has an electric field of 10^5V/m between the plates. If the charge on the capacitor plate is 1μC, then the force on each capacitor plate is

(1) 0.1N (2) 0.05N

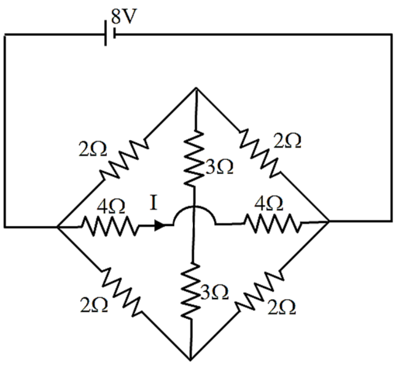
(3) 0.02N (4) 0.01N

44. In the circuit shown in the figure cell will deliver maximum power to the network if R (in Ω) is equal to



(1) 3 (2) 4 (3) 2 (4) 6

45. The value of I in the following circuit diagram will be :



(1) $\frac{3}{2}$A (2) $\frac{3}{4}$A

(3) $\frac{1}{2}$A (4) 1A

46. The far point of a near sighted person is 6.0 m from her eyes, and she wears contacts that enable her to see distant objects clearly. A tree is 18.0 m away and 2.0 m high. How high is the image formed by the contacts ?

(1) 0.1 m (2) 1.5 m

(3) 0.75 m (4) 0.50 m

47. A convex lens forms a real image on a screen placed at a distance 60 cm from the object . When the lens is shifted towards the screen by 20 cm , another image of the object is formed on the screen. The focal length of the lens is :

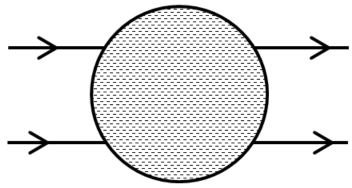
(1) 45 cm

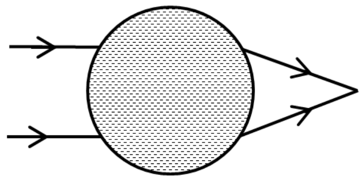
(2) 40 /3 cm

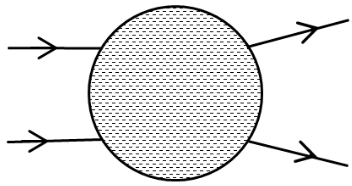
(3) 30 cm

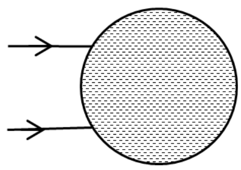
(4) 12 cm

48. A water drop in air refractes the light ray as :

(1) 

(2) 

(3) 

(4) 

49. In YDSE, phase difference between the light waves reaching third bright fringe from the central fringe will be (λ = 6000Å)

(1) Zero (2) 2π

(3) 4π (4) 6π

50. The maximum intensity of fringes in YDSE is I. If one of the slit is closed, then the intensity at that point is I_0. Which of the following relation is true ;

(1) $I = I_0$ (2) $I = 2I_0$

(3) $I = 4I_0$ (4) None

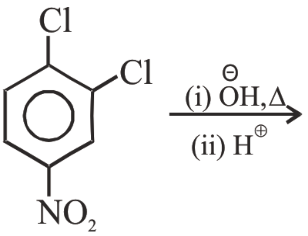
Topic : FRAR, NAR, ESR, S_NAE, S_NAR, SOLUTIONS, ELECTROCHEMISTRY (FROM GALVANIC CELL), SOLID STATE [BACK UNIT] :- SOLUTION

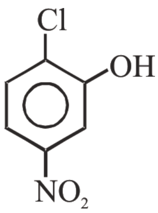
SECTION-A (CHEMISTRY)

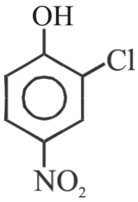
51. Major product obtained on reaction of phenol with NaOH and carbondioxide following by protonation, is :

(1) Benzoic acid (2) Salicyaldehyde

(3) Salicyclic acid (4) Phthalic acid

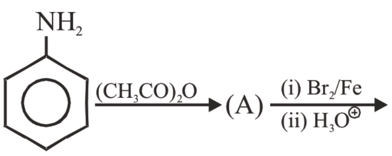
52.  P (Major), Based on above reaction, the major product would be :-

(1) 

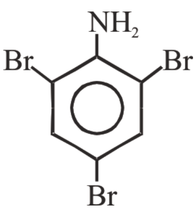
(2) 

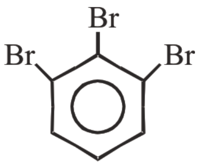
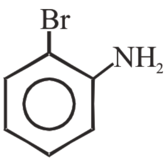
(3) Both in equal proportions

(4) No reaction

53.  (B) Major product

(B) in this reaction is :-

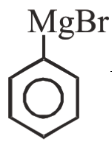
(1)  (2) 

(3)  (4) 

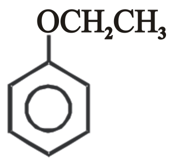
54. Chlorobenzene when condensed with chloral in the presence of conc. H_2SO_4 yields :

(1) Gammexane (2) DDT

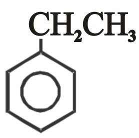
(3) TNB (4) C_6Cl_6

55.  $+ CH_3CH_2OH \longrightarrow ?$

(1)  $+ CH_3CH_2MgBr$

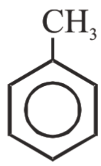
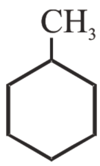
(2) 

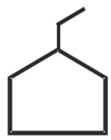
(3)  $+ CH_3CH_2OMgBr$

(4)  $+ MgBr(OH)$

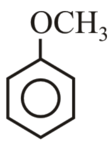
56.  $\xrightarrow[500°C]{Al_2O_3 + Cr_2O_3}$?

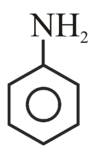
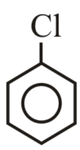
Major product is

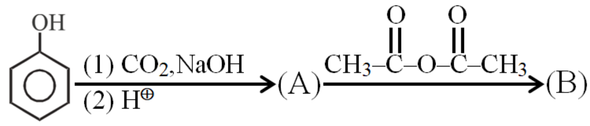
(1)  (2) 

(3)  (4) 

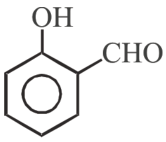
57. Which of following does not undergo Friedel Crafts reaction

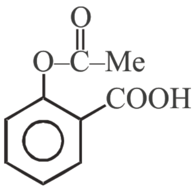
(1)  (2) 

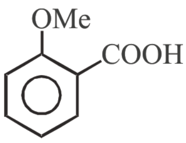
(3)  (4) 

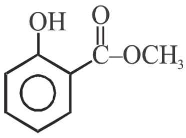
58. 

Product (B) is :

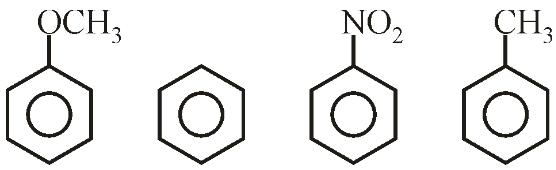
(1) 

(2) 

(3) 

(4) 

59. Increasing order of reactivity towards electrophilic substitution reaction for following compounds is :-



I II III IV

(1) III < II < IV < I

(2) II < III < IV < I

(3) III < II < I < IV

(4) III < I < II < IV

60. In electrophilic substitution reaction a chlorine substituent is :

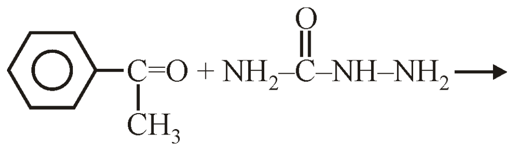
(1) Deactivating and meta directing

(2) Deactivating and o, p directing

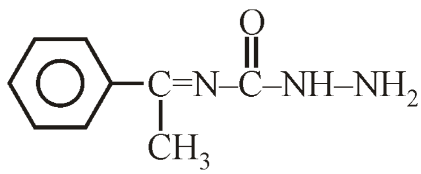
(3) Activating and meta directing

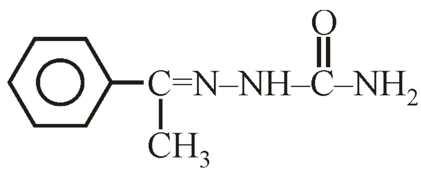
(4) Activating and o, p directing

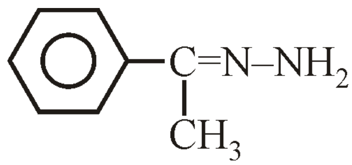
61. The main product of following reaction will be :-

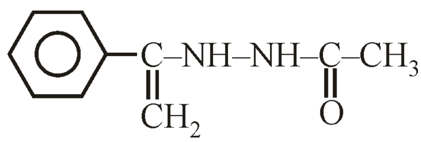


Product

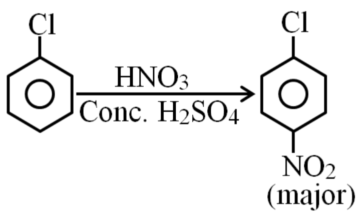
(1) 

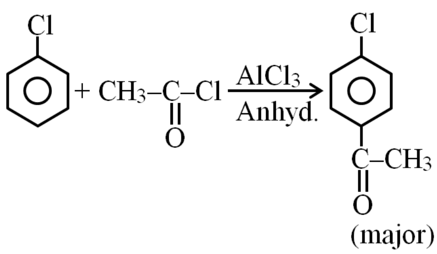
(2) 

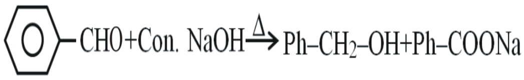
(3) 

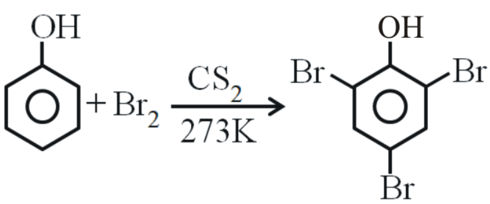
(4) 

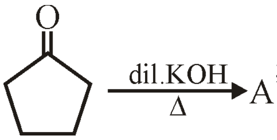
62. Which of the following is incorrect?

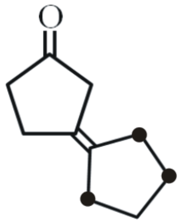
(1) 

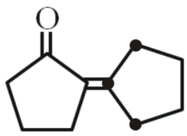
(2) 

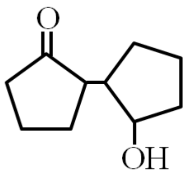
(3) 

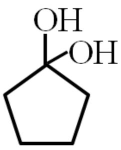
(4) 

63.  ; Product A is :-

(1) 

(2) 

(3) 

(4) 

64. Which of the following statement is false ?

(1) Units of atmospheric pressure and osmotic pressure are the same

(2) In reverse osmosis, solvent molecules move through a semipermeable membrane from a region of lower concentration of solute to a region of higher concentration.

(3) The value of molal depression constant depends on nature of solvent.

(4) Relative lowering of vapour pressure, is a dimensionless quantity.

65. The value of Henry's constant K_H is ....

(1) greater for gases with higher solubility

(2) greater for gases with lower solubility

(3) constant for all gases

(4) not related to the solubility of gases

66. On the basis of information given below mark the correct option.

Information :

(A) In bromoethane and chloroethane mixture intermolecular interactions of A–A and B–B type are nearly same as A – B type interactions.

(B) In ethanol and acetone mixture A–A or B–B type intermolecular interactions are stronger than A–B type interactions.

(C) In chloroform and acetone mixture A–A or B–B type intermolecular interactions are weaker than A–B type interactions.

(1) Solution (B) and (C) will follow Raoult's law

(2) Solution (A) will follow Raoult's law.

(3) Solution (B) will show negative deviation from Raoult's law.

(4) Solution (C) will show positive from Raoult's law.

67. One mole of a solute A is dissolved in a given volume of a solvent. The association of the solute take place as follows : $nA \rightleftharpoons A_n$

If α is the degree of association of A, the van't Hoff factor i is expressed as :

(1) $i = 1 - \alpha$

(2) $i = 1 + \frac{\alpha}{n}$

(3) $i = \frac{1 - \alpha + \frac{\alpha}{n}}{1}$

(4) $i = 1$

68. We have three aqueous solutions of NaCl labelled as A, B & C with concentrations 0.1 M, 0.01 M & 0.001 M respectively. The value of Van't Hoff factor for three solutions will be in the order.

(1) $i_A < i_B < i_C$

(2) $i_A > i_B > i_C$

(3) $i_A = i_B = i_C$

(4) $i_A < i_B > i_C$

69. Moles of Na_2SO_4 to be dissolved in 12 mol water to lower its vapour pressure by 10 mm Hg at a temperature at which vapour pressure of pure water is 50 mm is :

(1) 1.5 mol

(2) 2 mol

(3) 1 mol

(4) 3 mol

70. A 5% (w/V) solution of cane sugar (molecular weight = 342) is isotonic with 1% (w/V) solution of a substance X. The molecular weight of X is:

(1) 34.2

(2) 171.2

(3) 68.4

(4) 136.8

71. At 25°C, the vapour pressure of pure liquid A (mol. wt. = 40) is 100 torr, while that of pure liquid B is 40 torr, (mol.wt. = 80). The vapour pressure at 25°C of a solution containing 20 g of each A and B is :

(1) 80 torr

(2) 59.8 torr

(3) 68 torr

(4) 48 torr

72. Total vapour pressure of mixture of 1 mole $X(P_X^o = 150 \text{ torr})$ and 2 mole $Y(P_Y^o = 300 \text{ torr})$ is 240 torr. In this case :

(1) There is a negative deviation from Roult's law

(2) There is a positive deviation from Roult's law

(3) There is no deviation from Roult's law

(4) Cannot be decided

73. The freezing point of 1% aqueous solutions of calcium nitrate will be:

(1) 0°C (2) above 0°C

(3) 1°C (4) below 0°C

74. A pressure cooker reduces cooking time because :

(1) Heat is more evenly distributed

(2) b.pt. of water inside the cooker is increased

(3) The high pressure tenderises the food

(4) All of these

75. In which of the following crystals alternate tetrahedral voids are occupied ?

(1) NaCl (2) ZnS

(3) CaF_2 (4) Na_2O

76. The number of tetrahedral and octahedral holes in a hexagonal primitive unit cell are respectively :-

(1) 8, 4 (2) 6, 12

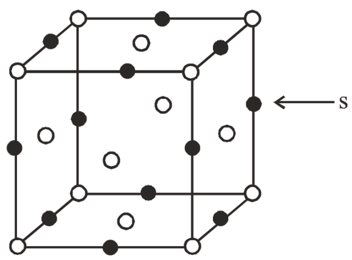
(3) 2, 1 (4) 12, 6

77. Lithium crystallizes as body centered cubic crystals. If the length of the side of unit cell is 350 pm, the atomic radius of lithium is :

(1) 151.5 pm (2) 123.7 pm

(3) 303.1 pm (4) 606.2 pm

78. In the figure given below, the site marked as S is a :



(1) tetrahedral void

(2) cubic void

(3) octahedral void

(4) None of the above

79. How many faraday are needed to reduce a mole of MnO_4^- to Mn^{2+} ?

(1) 4 (2) 5

(3) 3 (4) 2

80. For the following cell at 298 K

$Ni \mid Ni^{+2} \parallel Cu^{+2} \mid Cu$

calculate value of K_c [equilibrium constant]

Given : $E^o_{Ni^{+2}\mid Ni} = -0.25$ V

$E^o_{Cu^{+2}\mid Cu} = 0.34$ V

(1) 10^{20} (2) 10^{30}

(3) 10^{25} (4) 20

81. If the passage of current liberates H_2 at cathode and Cl_2 at anode then the solution is :-

(1) $CuSO_4(aq)$

(2) $CuCl_2(aq)$

(3) KCl(aq)

(4) AgCl(aq)

82. Three faradays of electricity was passed through an aqueous solution of iron(II) bromide. The mass of iron metal deposited at the cathode is (At. wt. of Fe is 56 g mol^{-1}):-

(1) 56 gm (2) 84 gm

(3) 112 gm (4) 168 gm

83. Standard reduction potentials for the following reactions are :-

$Fe^{3+} + 3e^- \longrightarrow Fe$; – 0.036 V

$Fe^{2+} + 2e^- \longrightarrow Fe$; – 0.44 V

What will be SEP for

$Fe^{3+} + e^- \longrightarrow Fe^{2+}$?

(1) 0.772 V

(2) – 0.404 V

(3) – 0.772 V

(4) 0.077 V

84. How many Cl^- ions are there around Na^+ ion in NaCl crystal ?

(1) 3

(2) 4

(3) 6

(4) 8

85. The decreasing size of the voids in a close packing is :-

(1) Cubic > Octahedral > Tetrahedral > Triangular

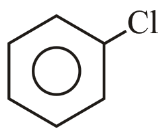
(2) Octahedral > Tetrahedral > Triangular > Cubic

(3) Tetrahedral > Triangular > Cubic > Octahedral

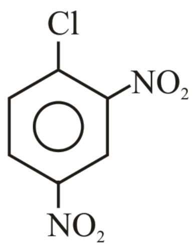
(4) Triangular > Cubic > Octahedral > Tetrahedral

SECTION-B (CHEMISTRY)

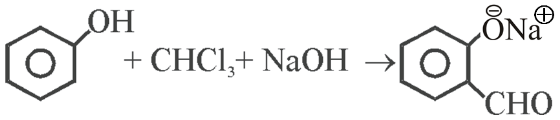
86. Which of the following compound is most reactive for aromatic nucleophilic substitution reaction.

(1) 

(2) 

(3) 

(4) 

87. 

The electrophile involved in the above reaction is :

(1) dichlorocarbene ($:CCl_2$)

(2) trichloromethyl anion $\left(\overset{\ominus}{C}Cl_3\right)$

(3) formyl cation $\left(\overset{\oplus}{C}HO\right)$

(4) dichloromethyl cation $\left(\overset{\oplus}{C}HCl_2\right)$

88. Which of the following compound undergo nucleophilic addition reaction with RMgX :-

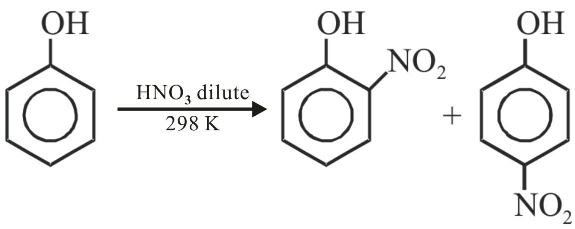
(1) $CH_3-\underset{\underset{O}{||}}{C}-CH_3$

(2) $CH_3-\underset{\underset{O}{||}}{C}-Cl$

(3) CH_3-Cl

(4) $CH_2=CH_2$

89. Which is incorrect for the following reaction?



(A) (B)

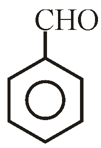
(1) This is an electrophilic substitution reaction.

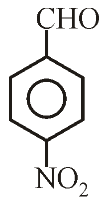
(2) A and B can be separated by vacuum distillation.

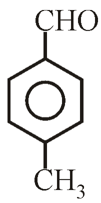
(3) B has higher Boiling Point than A

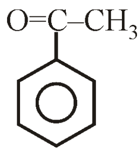
(4) B has intermolecular and A has intramolecular hydrogen bonding.

90. Which of the following is most reactive towards Nucleophilic addition reaction ?

(1) 

(2) 

(3) 

(4) 

91. Low concentration of oxygen in the blood and tissues of people living at high altitude is due to ....

(1) low temperature

(2) low atmospheric pressure

(3) high atmospheric pressure

(4) both low temperature and low atmospheric pressure

92. Normal saline solution is:

(1) 1N salt solution

(2) 0.9% mass/volume NaCl solution

(3) 1N NaCl solution

(4) Normal salt solution

93. One molal solution of a carboxylic acid in benzene shows the elevation of boiling point of 1.518 K. The degree of association for dimerization of the acid in benzene of (K_b for benzene = 2.53 K kg mol^{-1}) :

(1) 60%

(2) 70%

(3) 75%

(4) 80%

94. 0.1 M NaCl and 0.05M $BaCl_2$ solutions are separated by a semi-permeable membrane in a container. For this system, choose the correct answer : (assume complete dissociation)

(1) There is no movement of any solution across the membrane

(2) Water flows from $BaCl_2$ solution towards NaCl solution

(3) Water flows from NaCl solution towards $BaCl_2$ solution

(4) Osmotic pressure of 0.1 M NaCl is lower than the osmotic pressure of $BaCl_2$

95. Refractive index of a solid is observed to have the same value along all directions. It means nature of solid is–

(1) Anisotropic

(2) Isotropic

(3) Crystalline

(4) None of these

96. A metal crystallises in a cubic lattice containing a sequence of layers as ABABAB .... What percentage of voids are left in the lattice ?

(1) 72% (2) 48%

(3) 38% (4) 32%

97. In an electroplating experiment m grams of silver is deposited when 4 ampere of current flows for 2 minutes. The amount of silver deposited by 6 ampere of current flowing for 40 seconds will be :-

(1) 4m (2) m/2

(3) m/4 (4) 2m

98. For a cell $Cu\left|\underset{0.1\,M}{Cu^{2+}}\right|\left|\underset{0.001\,M}{Cu^{2+}}\right|Cu$;

The correct value of E_{cell} is :-

(1) $(-)\frac{RT}{F}\ln 0.1$

(2) $(-)\frac{RT}{F}\ln 10$

(3) $\frac{RT}{F}\ln 0.001$

(4) 0 V

99. Select the incorrect statement :

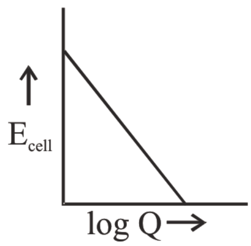
(1) Stoichiometry of crystal remains unaffected due to Schottky defect

(2) Frenkel defect is usually shown by ionic compounds having low coordination number

(3) F-centres generation is responsible factor for imparting the colour to the crystal

(4) Density of crystal always increases due to substitutional impurity defect

100. The plot of cell potential (E_{cell}) against logQ may be given as,



Which of the following is/are correct about the plot ?

(1) Slope of line = –0.059/n

(2) Intercept = E°_{cell}

(3) Slope of line = E°_{cell}

(4) Both (1) and (2)

Topic : STRUCTURAL ORGANIZATION IN ORGANISMS, ANIMAL TISSUE, BIOMOLECULES, MOLECULAR BASIS OF INHERITANCE : INTRODUCTION TO DNA STRUCTURE, LIFE CYCLE OF ANGIOSPERM (FROM FERTILIZATION TO APOMIXIS) [BACK UNIT] :- HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-I)

101. Match the Column I & II.

Column - I		Column - II	
A)	Cellulose	i)	Exoskeleton of arthropods
B)	Starch	ii)	Does not contain complex helices & can not hold I_2
C)	Chitin	iii)	With I_2 - blue in color
D)	Glycogen	iv)	Store house of energy in animal

(1) A – ii, B – iii, C – i, D – iv
(2) A – iii, B – ii, C – i, D – iv
(3) A – iii, B – ii, C – iv, D – i
(4) A – ii, B – iii, C – iv, D – i

102. Which of the following option consist of non-essential amino acids only :
(1) Valine, leucine, glycine, alanine
(2) Glycine, serine, proline, glutamic-acid
(3) Proline, aspartic acid, glutamic acid,methonine
(4) Cysteine, tyrosine, alanine, isoleucine

103. An example of protein with quaternary structure is :
(1) Myoglobin (2) Haemoglobin
(3) Keratin (4) All of these

104. Which of the following is not secondary metabolite ?
(1) Codeine (2) Lecithin
(3) Monoterpenes (4) Curcumin

105. Paper made from plant pulp is :-
(1) Galactose (2) Starch
(3) Cellulose (4) Glucosamine

106. Which of the following statement is true?
(1) A fatty acid has a carboxyl group attached to an R group.
(2) Palmitic acid has 16 carbon atoms including carboxyl carbon.
(3) Arachidonic acid has 20 carbon atoms including carboxyl carbon.
(4) All of the above

107. Match the ploidy of the following structures:

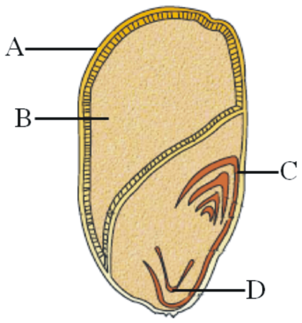
A.	Aleurone layer	(i)	2n
B.	Perisperm	(ii)	3n
C.	Embryo sac.	(iii)	n
D.	Scutellum	(iv)	2n

(1) A-ii, B-iii, C-i, D-iv
(2) A-iii, B-iv, C-ii, D-i
(3) A-iv, B-iii, C-i, D-ii
(4) A-ii, B-iv, C-iii, D-i

108. Which one of the following is an example of Non-endospermic seed ?
(1) Wheat
(2) Coconut
(3) Ground nut
(4) Castor

109. The drinking portion (Coconut water) is ____(A)____and edible portion (white kernel) is the ___(B)___ in coconut.
(1) A-Cellular endosperm, B-Nuclear endosperm
(2) A-Nuclear endosperm, B-Cellular endosperm
(3) A-Starchy endosperm, B-Nuclear endosperm
(4) A-Oily endosperm, B-Starchy endosperm

110. Which of the following set represents the correct labelling of A, B, C and D with respect to the given diagram?



(1) A-Pericarp, B-Endosperm, C-Coleoptile, D-Coleorhiza

(2) A-Endosperm, B-Pericarp, C-Coleorhiza, D-Coleoptile

(3) A-Pericarp, B-Endosperm, C-Coleorhiza, D- Coleoptile

(4) A-Perisperm, B-Endocarp, C-Scutellum, D-Radicle

111. The production of seeds without Fertilization is called :-

(1) Parthenogenesis (2) Partheno carpy

(3) Apomixis (4) Polyembryony

112. A group of similar cells along with intercellular substance called tissue in?

(1) Virus

(2) Unicellular eukaryote

(3) Unicellular prokaryote

(4) Multicellular animals

113. **Assertion :** Compound epithelium is made up of more than one layer (Multi-layered) of cell's

Reason : Their main function is to provide protection against chemical and mechanical stress.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

114. Which of the following structure does not contain ciliated epithelium ?

(1) PCT of Nephron

(2) Fallopian tube

(3) Trachea and Bronchioles

(4) Collecting duct of Nephron

115. Epithelium which is found in wall of Alveoli is made up of –

(1) Loosely arranged cells

(2) Compactly packed single layer of cells

(3) Compactly packed multiple layer of cells

(4) Non-cellular layer of hyaluronic acid

116. Read the following statements and select the correct option.

Statement-I : Cell are compactly packed in the epithelial tissues with little intercellular matrix.

Statement-II : The cells secrets fibers of structural protein in all the connective tissues except blood.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

117. Match the columns and choose correct option :

	Column-I		Column-II
(i)	Ciliated epithelium	(a)	Airsacs of lungs
(ii)	Cuboidal epithelium	(b)	Fallopian tubes
(iii)	Cuboidal cell with microvill	(c)	PCT of nephron
(iv)	Ciliated columnar cells	(d)	Bronchioles
(v)	Squamous epithelium	(e)	Ducts of glands

	i	ii	iii	iv	v
(1)	b	a	b	c	d
(2)	d	e	c	a	b
(3)	d	e	c	b	a
(4)	d	b	c	a	e

118. All structures are having compound epithelium except one-
(1) Dry surface of skin
(2) Moist surface of buccal cavity
(3) Pharynx
(4) Bronchiole

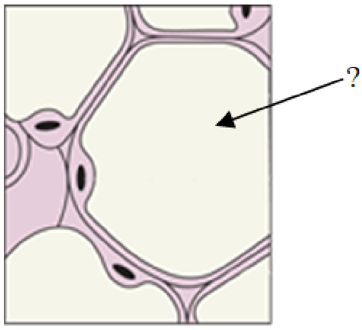
119. On the basis of percent body weight, muscles are :-
(1) 20-30 %
(2) 30-40 %
(3) 40-50 %
(4) 15-20 %

120. **Assertion :** Exocrine gland secrete Mucus, Saliva, Earwax, Oil, Milk digestive enzyme and other products.
Reason : Endocrine gland do not have duct and these gland secret hormone.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

121. In which structure, microvilli are present ?
(1) PCT
(2) Wall of blood vessels
(3) Fallopian tubes
(4) Bronchioles

122. Junction, which perform cementing to keep neighbouring cells together is :-
(1) Interdigitation
(2) Gap Junction
(3) Adhering junction
(4) Tight junction

123. In the following diagram Arrow indicate which structure ? (Loose -connective Tissu)



(1) Fat storage area (2) Nucleus
(3) Plasma membrane (4) Matrix

124. The most abundent & widely distributed tissue is __A__ & function of __B__ & __C__.
Find out the correct ans for A, B & C
(1) A=Epithelium, B=Linking, C=Supporting
(2) A=Connective, B=Lining, C=Covering
(3) A=Epithelium, B-Lining, C=Covering
(4) A=Connective, B=Linking, C=Supporting

125. ____A____ is a example of simple squamous epithelium which is related with __B__ .
(1) A–Duct of glands; B–Circulation
(2) A–Fallopian tube; B–Reproduction
(3) A–Proximal convoluted tubule; B–Excretion
(4) A–Air sac of lungs; B–Respiration

126. **Assertion :** Blood is a type of specialised connective tissue.
Reason : In Blood, cells do not secrete fibres of structural proteins, called collagen or elastin.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

127. Which of the following statement is/are incorrect ?

(A) Cell junctions are only found in epithelial tissue

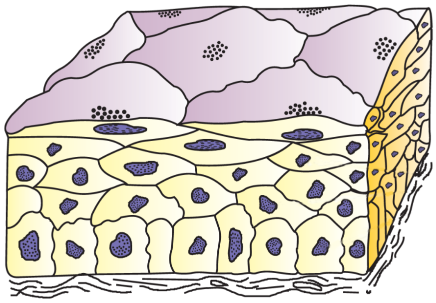
(B) Cell junctions provide structural and functional links between individual cells

(C) Adhering junctions perform cementing to facilitate communication between neighbouring cells

(1) Only B (2) Only A and C

(3) All A, B and C (4) Only C

128.



Firstly identify the figure and choose correct option according to related tissue.

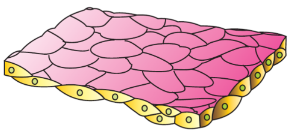
(1) It has main role in absorption and secretion

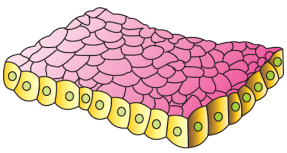
(2) It form the lining of body cavities

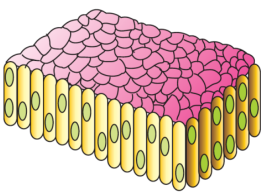
(3) It is found in dry surface of skin

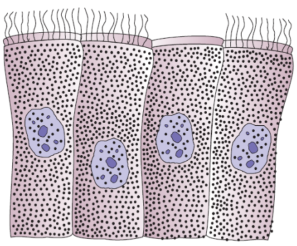
(4) It is found in air sacs of lung.

129. Tissue which forms a diffusion boundary is :-

(1) 

(2) 

(3) 

(4) 

130. Match the following column and select the correct option :

Column-I		Column-II	
(A)	Squamous epithelium	i	Move particles or mucus in specific direction
(B)	Columnar epithelium	ii	Nuclei are located at the base
(C)	Ciliated epithelium	iii	Cells with irregular boundaries
(D)	Cuboidal epithelium	iv	Main function is secretion and absorption

(1) A – i, B – ii, C – iv, D – iii

(2) A – iii, B – ii, C – i, D – iv

(3) A – ii, B – i, C – iii, D – iv

(4) A – iv, B – i, C – iii, D – i

131. Which of the following statement is correct?

(1) On the basis of structural modification of the cells, simple epithelium is divided into three types.

(2) If the columnar or cuboidal cells bear cilia on their basal surface, they are called ciliated epithelium.

(3) The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has cilia.

(4) Cuboidal epithelium is involved in functions like forming a diffusion boundary.

132. Select the correct statements :

(A) Human body is composed of billions of cells

(B) Tissue is group of similar cells along with intra cellular substances

(C) All complex animals consist only four basic type of tissues

(D) Tissues are organised in specific proportion and pattern to form organ

(1) Only C and D (2) Only B, C and D

(3) Only A, C and D (4) All

133. Choose incorrect statement :

(a) All gland are multicellular glands except goblet cells and paneth cells.

(b) Epididymis have Brush bordered epithelium.

(c) Inner lining of duct of salivary gland is made up of more than one layer of cells.

(d) Tendon attach skeletal muscle to bone is a example of loose connective tissue.

(1) (a) and (b)   (2) (b) and (d)

(3) (c) and (d)   (4) (b), (c) and (d)

134. Fill in the blanks and choose the correct option

(A) The ___A___ of the cells varies according to their ___B___.

(B) The squamous epithelium is made of a single thin layer of flattened cells with ___C___ boundaries. They are found in the walls of ___D___ and ___E___.

	A	B	C	D	E
1.	Structure	Function	Regular	Blood vessels	Air sac of lungs
2.	Function	Structure	Regular	Blood vessels	Duct of glands
3.	Structure	Function	Irregular	Blood vessels	Air sac of lungs
4.	Function	Structure	Irregular	Blood vessels	Air sac of lungs

135. **Assertion :** The rupture of RBC is associated with toxin haemozoin.

Reason : Plasmodium attack RBC and cause fever.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

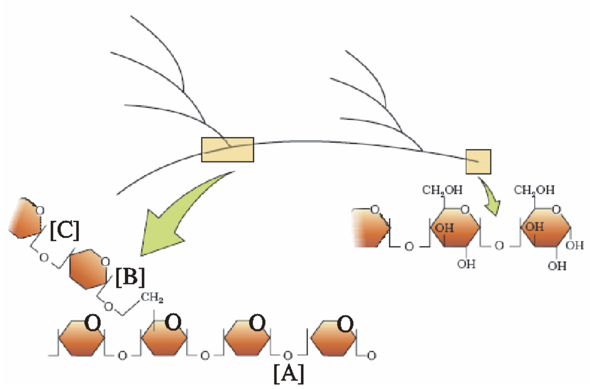
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

SECTION-B (BIOLOGY-I)

136.



In the above diagrammatic representation of a portion of glycogen, glycosidic bonds represented by A, B and C are :-

	A	B	C
(1)	α-1'-4"	α-1'-4"	β-1'-4"
(2)	β-1'-6"	β-1'-4"	β-1'-4"
(3)	α-1'-6"	α-1'-4"	α-1'-4"
(4)	α-1'-4"	α-1'-6"	α-1'-4"

137. Find out the correct match from the column-I and column-II.

	Column-I		Column-II
(a)	Pectin	(i)	Hormone
(b)	Trypsin	(ii)	Steroid
(c)	GLUT-4	(iii)	Polysaccharide
(d)	Insulin	(iv)	Enzyme
(e)	Cholesterol	(v)	Transporter protein

(1) (a)-(iii), (b)-(iv), (c)-(v), (d)-(i), (e)-(ii)

(2) (a)-(v), (b)-(iii), (c)-(i), (d)-(ii), (e)-(iv)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii), (e)-(v)

(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(v), (e)-(iii)

138. The R group in serine is :-

(1) Methyl

(2) Oxygen

(3) Hydrogen

(4) Hydroxy methyl

139. After fertilization the following structure of ovule change in to?

Structure		Change into	
A.	Outer Integument	(i)	Tegmen
B.	Nucellus	(ii)	Scar
C.	Hilum	(iii)	Perisperm
D.	Inner Integument	(iv)	Testa

(1) A-iii, B-iv, C-ii, D-i

(2) A-iv, B-iii, C-ii, D-i

(3) A-iv, B-iii, C-i, D-ii

(4) A-ii, B-i, C-iii, D-iv

140. What will be the ploidy of endosperm of a seed produced after crossing hexaploid female plant with tetraploid male plant ?

(1) Hexaploid

(2) Pentaploid

(3) Octaploid

(4) Triploid

141. A supportive tissue in human outer ear joints & tip of nose is :-

(1) Bones

(2) Cartilage

(3) Muscles

(4) Adipose tissue

142. Which statement is not true in respect of tendon ?

(1) It connects skeletal muscles to bones

(2) It is made up of dense irregular connective tissue

(3) It exhibits regular orientation of fibres

(4) It is made up of dense regular connective tissue

143. **Assertion :** Adipose tissue is the type of loose connective tissue located mainly beneath the skin.
Reason : The excess of nutrients which are not used immediately are converted into fats and are stored in this tissue.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

144. Which type of tissue have communication junctions between adjacent cells and striation pattern?

(1) Cardiac muscle fibre

(2) Connective tissue

(3) Skeletal muscle fibre

(4) 1 and 3 both (Cardiac muscle fibre and skeletal muscle fibre)

145. Read the following statements and select the correct option :

Statement-I : The columnar epithelium is composed of single layer of tall and slender cells.

Statement-II : Tight junctions help to stop substances from leaking across a tissue.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

146. Read the following statements :-

(a) Dense irragular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently.

(b) Cartilage, bones and blood are various types of areolar tissues.

(c) The intercelluar material of cartilage is fluid, non pliable and sensitive to compression.

(1) a, b is correct and c is incorrect

(2) a, c is correct and b is incorrect

(3) b, c is correct and a is incorrect

(4) b, c is incorrect and a is correct

147. Which feature is not related to bone ?

(1) Bone has solid and pliable ground substance

(2) Matrix rich in calcium salts and collagen fibres

(3) Provides structural frame to the body

(4) Some bones are site of production of blood cells.

148. Fibres & fibroblast are compactly packed in which type of tissue?

(1) Areolar tissue

(2) Adipose tissue

(3) Dense regular tissue

(4) Epithelium tissue

149. **Statement-I :** Bones have hard & pliable ground substance.

Statement-II : Cartilage has solid & non-pliable matrix.

(1) Statement I & II both are correct.

(2) Statement I is correct & II is incorrect

(3) Statement I is incorrect & II is correct

(4) Both statement I & II is incorrect

150. Read the following statements and choose the correct option

Statement 1: The body of a simple organism like *Hydra* is made of different types of cells and the number of cells in each type can be in millions.

Statement 2: When two or more organs perform a common function only through their physical interaction, they together form organ system, e.g., digestive system, respiratory system, etc.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement 1 is correct

(4) Only statement 2 is correct

Topic : STRUCTURAL ORGANIZATION IN ORGANISMS, ANIMAL TISSUE, BIOMOLECULES, MOLECULAR BASIS OF INHERITANCE : INTRODUCTION TO DNA STRUCTURE, LIFE CYCLE OF ANGIOSPERM (FROM FERTILIZATION TO APOMIXIS) [BACK UNIT] :- HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-II)

151. Read the following statements-

(A) The rupture of RBCs is associated with release of a toxic substance, haemozoin, which is responsible for the chill and high fever recurring every one to two days.

(B) The new-born receives some antibodies from their mother, through the placenta during pregnancy.

(C) Lymph nodes serve to trap the micro-organisms or other antigens, which happen to get into the blood and tissue fluid.

(D) After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of large intestine and appendix.

(E) MRI uses strong magnetic fields and ionising radiations to accurately detect pathological and physiological changes in the living tissue.

How many of the above statements are **not correct**?

(1) Three (2) Four (3) Five (4) Two

152. Match these columns :

	Column-I		Column-I
A.	Acid in stomach and tears from eye	i.	Cellular immunity
B.	Antibody	ii.	Physical barrier
C.	T-lymphocyte	iii.	Humoral immunity
D.	Mucous Coating	iv.	Physiological barrier

(1) A-iv, B-ii, C-iii, D-i

(2) A-ii, B-iv, C-i, D-iii

(3) A-iv, B-iii, C-i, D-ii

(4) A-iv, B-i, C-ii, D-iii

153. Given below are two statements –

Statement-I : Injection of dead/Inactivated pathogen causes active immunity.

Statement-II : Antibody represented H_2L_4 structure.

In the light of the above statements choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are correct

154. Which of the following pair is not matched correctly ?

(1) *Sickle cell anemia* ⟶ Autosome linked recessive

(2) *Haemophilia* ⟶ Sex linked recessive

(3) *Colour blindness* ⟶ Autosome linked recessive

(4) *Phenylketonuria* ⟶ Autosomal recessive

155. What is the genetic disorder in which females are sterile as ovaries are rudimentary besides other features including lack of other secondary sexual characters ?

(1) Turner's syndrome

(2) Klinefelter's syndrome

(3) Edward syndrome

(4) Down's syndrome

156. *Wuchereria* infects :-

(1) Alveoli of lungs

(2) Lymphatic vessels

(3) Nose and respiratory passage

(4) Intestine

157. Among the following which does not cause ringworm disease in humans?

(1) Microsporum (2) Macrosporum

(3) Epidermophyton (4) Trichophyton

158. Match the correct:-

(a)		(i)	*Cannabis sativa*
(b)		(ii)	*Datura*
(c)		(iii)	*Opium poppy*

(1) (a) - (iii), (b) - (ii), (c) - (i)

(2) (a) - (iii), (b) - (i), (c) - (ii)

(3) (a) - (i), (b) - (iii), (c) - (ii)

(4) (a) - (i), (b) - (ii), (c) - (iii)

159. **Statement-I :** Because of the addictive nature of alcohol and drug and their perceived benefit like relief from stress, a person may try taking these in the face of peer pressure, examination-related stress.

Statement-II : Some of us are sensitive to some particle in environment and suddenly started sneezing because of allergy to mites etc.

(1) Statement-I is correct II is false.

(2) Statement-II is correct I is false.

(3) Statement-I & II both are correct.

(4) Statement-I & II both are incorrect.

160. Opportunistic infections during AIDS include -

(1) Tuberculosis

(2) Pneumonia

(3) Oropharyngeal candidiasis

(4) All of these

161. Which of the follow is the property of cancer cell -

(1) Contact inhibition

(2) Presume of C-onc

(3) Inactivated protooncogene

(4) Maintain telomerase activity

162. Most common cancer in females in world wide-

(1) Cervical cancer

(2) Breast cancer

(3) Lung cancer

(4) Adenocarcinoma

163. Primary Immunodeficiency is -

(1) AIDS

(2) SCID

(3) Slim disease

(4) All of these

164. Which of the following STI is caused by bacteria -

(1) Genital warts

(2) Hepatitis B

(3) Chlamydiasis

(4) Genital herpes

165. In Burkitt's lymphoma reciprocal translocation occurs between -

(1) 9 and 22

(2) 8 and 14

(3) 5 and 21

(4) 13 and 17

166. Which of the following disorder not include in trisomy-

(1) Downs syndrome

(2) Edward syndrome

(3) Cat cry syndrome

(4) Patau syndrome

167. Complete set of chromosome increase during -

(1) Aneuploidy

(2) Polyploidy

(3) Gene disorder

(4) Euploidy

168. HAART treatment include

(1) Protease Inhibitors

(2) Reverse transcriptase inhibitors

(3) Integrase Inhibitors

(4) All of these

169. Which of the following is psychotropic drug.

(1) Stramonium

(2) Atropine

(3) Morphine

(4) LSD

170. Which of the following not used during chemotherapy

(1) Vinblastine drug

(2) Taxol drug

(3) Co - 60

(4) All of these

171. Select the correct option for carcinogen and cancer-

(1) DES ⟶ Oral cancer

(2) Benzopyrene ⟶ Brain tumor

(3) UV Rays ⟶ Xeroderma Pigmentosum

(4) Kangri cancer ⟶ Prostate cancer

172. Select the correct option for carcinoma

(1) Most rare type of cancer

(2) Receprocal translocation is responsible for carcinoma

(3) Cancer of epithelial tissue

(4) Include CML and Burkitt's cancer

173. Viral disease which affect upper respiratory tract is -

(1) Asthma

(2) Pneumonia

(3) Common cold

(4) Emphysema

174. Adenine + Ribose + Phosphate form :-

(1) Adenosine

(2) Deoxyadenylic acid

(3) Adenylic acid

(4) Deoxyadenosine

175. DNA and RNA molecules differ from each other with respect to ?

(1) A purine base

(2) A pyrimidine base

(3) Pentose sugar

(4) Both (2) & (3)

176. What is the nature of 2 strands of a DNA duplex ?

(1) Identical and complimentary

(2) Anti-parallel and complimentary

(3) Dissimilar and non-complimentary

(4) Anti-parallel and non complimentary

177. Short DNA segment has 80 Thymine and 90 Guanine bases. The total number of nucleotides in DNA segment are :-
(1) 160
(2) 240
(3) 180
(4) 340

178. **Assertion:** In DNA base pairing generates approximately uniform distance between the two strands of the helix.
Reason: During base pairing, always a purine comes opposite to a pyrimidine.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

179. Backbone of DNA structure is made up of -
(1) Sugar only
(2) Phosphate only
(3) Sugar and Phosphate
(4) Sugar and Nitrogenous base.

180. Which bond is not found in a nucleotide ?
(1) N-glycosidic bond
(2) Phosphoester bond
(3) Phospodiester bond
(4) None of the above

181. If sequence of one strand of DNA is written as follows :
5'ATGC3'
Sequence of it's complementary strand will be
(1) 5'TACG3'
(2) 5'GCAT3'
(3) 5'ATGC3'
(4) 5'CGTA3'

182. If a DNA has 4000 bp, then how many phosphodiester bonds present ?
(1) 3998 (2) 7998
(3) 4000 (4) 8000

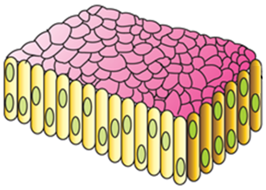
183. Length of pitch of the DNA helix is :
(1) 3.4 nm (2) 10 nm
(3) 34 nm (4) 0.34 nm

184. Main Hallmark feature of DNA double helix model is :
(1) Antiparallel strands of DNA
(2) Complementary base pairing
(3) Double stranded
(4) Helical structure

185. Double Helix model of DNA proposed by :
(1) Watson and Crick
(2) Friedrich Meischer
(3) Wilkens and Franklin
(4) Altman

SECTION-B (BIOLOGY-II)

186. Structure which is given below diagram is related with which type of epithelium and present in which part of body?



(1) A = Simple squamous epithelium ;
B = Blood vessels

(2) A = Columnar epithelium; B = Stomach

(3) A = Stratified columnar epithelium ;
B = Pharynx

(4) A = Cuboidal epithelium;
B = Tubular part of nephron in kidney

187. Select the odd one with respect to tissue it contain –

(1) Outer Ear joint

(2) Between adjacent bones of vertebral column

(3) Tip of Nose

(4) Tendons

188. Match the following column?

	Column-I		Column-II
(A)	Dense irregular connective tissue	(i)	Solid, Pliable and resist compression
(B)	Areolar loose connective tissue	(ii)	Fibroblasts and mostly collagen fibre oriented differently
(C)	Specialized connective tissue (cartilage)	(iii)	Blood
(D)	Fluid connective tissue	(iv)	Cells and fibre loosely arrange and present beneath the skin

(1) A–(i); B–(iv); C–(iii); D–(ii)

(2) A–(iv); B–(i); C–(iii); D–(ii)

(3) A–(ii); B–(iv); C–(i); D–(iii)

(4) A–(iii); B–(i); C–(iv); D–(ii)

189. **Assertion (A) :–** Common examples of allergens are mites in dust, pollens, animal dander etc.

Reason (R) :– Substances to which exaggerated immune response is shown are called allergens.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

190. Match column I with column II

	Column-I		Column-II
(i)	Homo graft	(a)	Active immunity
(ii)	Bone marrow	(b)	Non-specific
(iii)	Innate immunity	(c)	Primary lymphoid organ
(iv)	Small pox vaccine	(d)	Same species

(1) (i)-(a),(ii)-(b),(iii)-(c),(iv)-(d)

(2) (i)-(b),(ii)-(a),(iii)-(d),(iv)-(c)

(3) (i)-(d),(ii)-(c),(iii)-(b),(iv)-(a)

(4) (i)-(c),(ii)-(d),(iii)-(a),(iv)-(b)

191. **Statement-I :** Aquired immunity is pathogen non-specific.

Statement-II : Innate immunity is specific type of defence.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

192. Match the correct column.

	Column-I		Column-II
A.	Opioids	i.	Treatment of insomnia
B.	Cannabinoids	ii.	Coca alkaloid
C.	Cocaine	iii.	Chars and Ganja
D.	Barbiturates	iv.	Smack

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-iii, C-ii, D-i

(3) A-ii, B-i, C-iii, D-iv

(4) A-iii, B-iv, C-i, D-ii

193. Select incorrect for turner's syndrome-

(1) 44 + xo

(2) Rudimentary Ovary

(3) Chromosomal aneuploidy

(4) Mentally retard

194. Vision producing drug is called

(1) Psychotropic drug

(2) Depressant drug

(3) Mood altering drug

(4) Psychedelic drug

195. Which you will not get on hydrolysis of a nucleoside ?

(1) Purine

(2) Pyrimidine

(3) Pentose sugar

(4) Phosphate group

196. The plane of one base pair stacks over the other in double helix. This provides :-

(1) Antiparallel nature to DNA double helix

(2) Uniform length in all DNA

(3) Unifrom width throughout DNA

(4) Additional stability to DNA

197. **Assertion :-** DNA is more chemically stable than RNA.

Reason :- In DNA every nucleotide residue has an 'H' group at 2' position in the deoxyribose sugar while RNA has 2' OH group.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

198. In a dsDNA, if 20% Adenine is present, then find out the percentage of 5-methyluracil in this DNA segment.

(1) 0%

(2) 20%

(3) 30%

(4) 60%

199. Total number of base pairs present in a DNA segment with length 68 nm :

(1) 340 bp

(2) 200 bp

(3) 680 bp

(4) 10 bp

200. Match List-I with List-II :

	List-I		**List-II**
(a)	Bacteriophage $\phi \times 174$	i.	48502 base pairs
(b)	Bacteriophage lambda	ii.	5386 nucleotides
(c)	Escherichia coli	iii.	3.3×10^9 base pairs
(d)	Haploid content of human DNA	iv.	4.6×10^6 base pairs

Choose the **correct** answer from the options given below :

(1) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

(2) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

(3) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

(4) (a) - (i), (b) - (ii), (c) - (iv), (d) - (iii)

QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)

ENTHUSE ADV. COURSE : MED

MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
05 16-07-2023	SYLLABUS COVERED FROM 19-06-2023 TO 15-07-2023	P	CURRENT ELECTRICITY
		C	ELECTRO CHEMISTRY
		B	LIFE CYCLE OF ANGIOSPERM

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK